PATIENT CARE THEORY 2

UNIT 8: Genitourinary/Renal/Reproductive

Marilyn Niffin BSc, ACP
Winter 2020, 2021, 2023

Learning Objectives

- Differentiate the clinical manifestations of various renal disorders
- Identify situations and/or conditions for which an emergency disconnect from home dialysis may be required.
- Identify the types of soft tissue injuries and the manifestations that may occur to the external/internal male genitalia.
- Identify the location and explain the function of the female reproductive structures:

Genitourinary Disorders

- UTI
  - pyelonephritis,
  - Hydronephrosis
- Urologic Stone disease
- Acute kidney Injury (formerly Acute Renal Failure)
- Chronic kidney disease (formerly chronic renal Failure)
  - Hemodialysis Emergency Home Disconnect
- Foreign body
- Urinary retention/incontinence
- Foley Catheters (types and care)

Kidney infection

fluid accumulation in kidneys

Gynecological/Reproductive

- Male Reproductive
  - Testicular torsion
  - Epididymitis
  - Orchitis
  - Prostate Enlargement
- Female gynecological:
  - Dysmenorrhea/amenorrhea
  - PID
  - Ovarian cyst
  - endometriosis

Types of pain

- Associated with GU disorders
  - Visceral – hollow structures ; ureters, bladder, urethra, vas deferens, epididymis (males)
    - Characterized by achy, cramping, deep pain, poorly localized
  - Referred – pain felt in a location other than its site of origin

Ectopic pregnancies and ovarian cysts

UTI (Urinary Tract Infection)

Definition

- bacteriuria (bacteria in urine) in the presence of symptoms
- Usually develop in the lower urinary tract
  - Can move to the upper urinary tract when left untreated
  - Pyelonephritis
  - urosepsis

Urinary Tract Infection (UTI)

Male

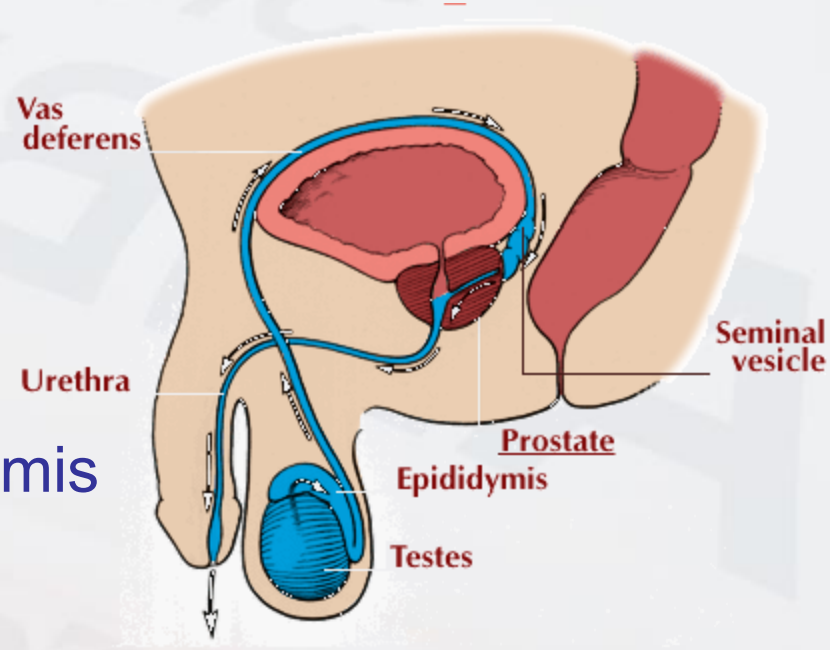
- bacterial colonization of the urinary tract
  - ie, kidney, ureter, bladder
- incidence of UTI is low in men
  - Therefore, anatomical abnormalities should be considered
- most frequent chief complaint with UTI is dysuria

obstruction due to inflammation or particulate (pus, etc.)

Urinary Tract Infection (UTI)

Male

- Urethritis
  - Inflammation of the urethra
- Epididymitis
  - Inflammation of the epididymis
- Prostatitis
  - Inflammtion of the prostate
- Orchitis
  - Inflammation of the testicle(s)
- pyelonephritis
  - Kidney infection



Urinary Tract Infection (UTI)

Female

- affects an estimated 20% of women at some time during their lifetimes
- Uncomplicated UTI involves the urinary bladder (cystitis)
- Etiology: most cases = enteric coliform bacteria (eg, *Escherichia coli*)
- inhabits the periurethral vagina

Urinary Tract Infection (UTI)

- Frequent and complete voiding has been associated with a reduction in the incidence of UTI
  - Does anyone need a bathroom break?

Common Signs and Symptoms

- Dysuria (painful urination); visceral in nature
  - Burning on urination
- Increased frequency and/or urgency to void
- Difficulty urinating
- Cloudy urine
- Foul smelling
- Pain on palpation of the lower abdomen (central/ above symphysis pubis)
- Note: Elderly patients may not present with these "classic symptoms"

altered LOA often first symptom

Urinary Tract Infection (UTI) - Pyelonephritis

Definition

- an infection of the renal pelvis
- Ascending Urinary Tract Infection: Infection which has began in the urinary tract which has ascended to the renal pelvis.
  - Dysuria, increased frequency, Nocturia, hesitancy (uneasy or interrupted), dark, cloudy urine
  - N/V
  - flank pain, fever, chills
- Risk of sepsis

Urinalysis

- Combines physical (visual), chemical (dipstick) and microscopic tests
  - Measures byproducts of normal metabolism, cells, cellular fragments and bacteria
  - Can be used to diagnose things like
    - Pregnancy, diabetes, UTI, kidney stones, cancer, liver disease
- Culture – a diagnostic lab test performed to detect the presence of bacteria in the urine (24-48 hrs for results)

foam = proteins

Urinalysis

- Visual – colour, consistency, precipitate
- Chemical (Dipstick) – pH, proteins, glucose, ketones, bilirubin, infection (nitrites or leukocytes), blood
- Microscopic – crystals, leukocytes, erythrocytes, bacteria, yeast, epithelial cells, crystals etc.
- Sample Urinalysis Report
- There will be “normal” ranges
- Abnormalities need to be interpreted in the broader clinical context -
  - Epithelial cells often indicate contamination
  - Presence of bacteria AND WBC’s is more meaningful than bacteria alone

Hydronephrosis

- swelling of a kidney due to a build-up of urine
  - Kidney is unable to drain
    - Obstruction, UTI (blood clots), Renal calculi
- If untreated, can lead to permanent kidney damage
- Typically affects only 1 kidney

Urologic Stone Disease

- wide variety of pathological processes, intrinsic and extrinsic to the urinary system, can cause obstruction
  - renal calculi (kidney stones) are the most common cause in middle aged men
  - In men over age 60 years, urinary obstruction is most commonly due to prostatic hypertrophy…occasionally due to prostate cancer
  - gynecologic surgery, pregnancy, and cancers of pelvic organs are important etiologies of obstruction in women
- Urinary obstruction is a common cause of acute and chronic renal failure

Urologic stone disease

- Often begin with vague symptoms of discomfort
  - Flank or groin area
  - Increases intensely with stone movement
  - May or may not have difficulty urinating (dependent upon stone location)
  - May have intense pain
  - (79) How Kidney Stones Are Formed Animation - Renal Calculi Causes & Symptoms Video - Blocked Urine Flow - YouTube
- Consider history (including family Hx); S&S of infection

dysuria only if stone betlow bladder, preventing emptying

Urologic Stone Disease

Renal colic

- Assess OPQRST – AS/S PN
- SpO2, Cardiac monitor
- Position of comfort (may have difficulty staying still)
- IV access
- Pain management (see ALS PCS)
  - Ketorolac is preferred

Foreign Body

- Any foreign body lodged somewhere in the GI or GU tract (intentionally or unintentionally)
  - Often sexual or erotic in nature
  - May require surgical removal (minimally invasive techniques used whenever possible)
  - DO NOT attempt to remove the object. Transport for further assessment

Trauma to the Genitourinary System

- Kidneys, ureters, bladder
- Kidneys well protected by the rib cage in adults
  - Exposed kidneys in children as they grow in height faster than the growth of the ribs
- 3-10% of patients with multiple trauma have GU involvement;
- 10-15% of trauma patients with abdominal injuries have GU involvement.
- Renal injuries constitute 45% of all GU injuries;
- ureteral injuries constitute 6%.
- See link posted on Blackboard for complete review of GU trauma

if patient has a full bladder and a seatbelt on, it can rupture during an MVC

GYNECOLOGICAL/REPRODUCTIVE

MALE GENITAL PROBLEMS

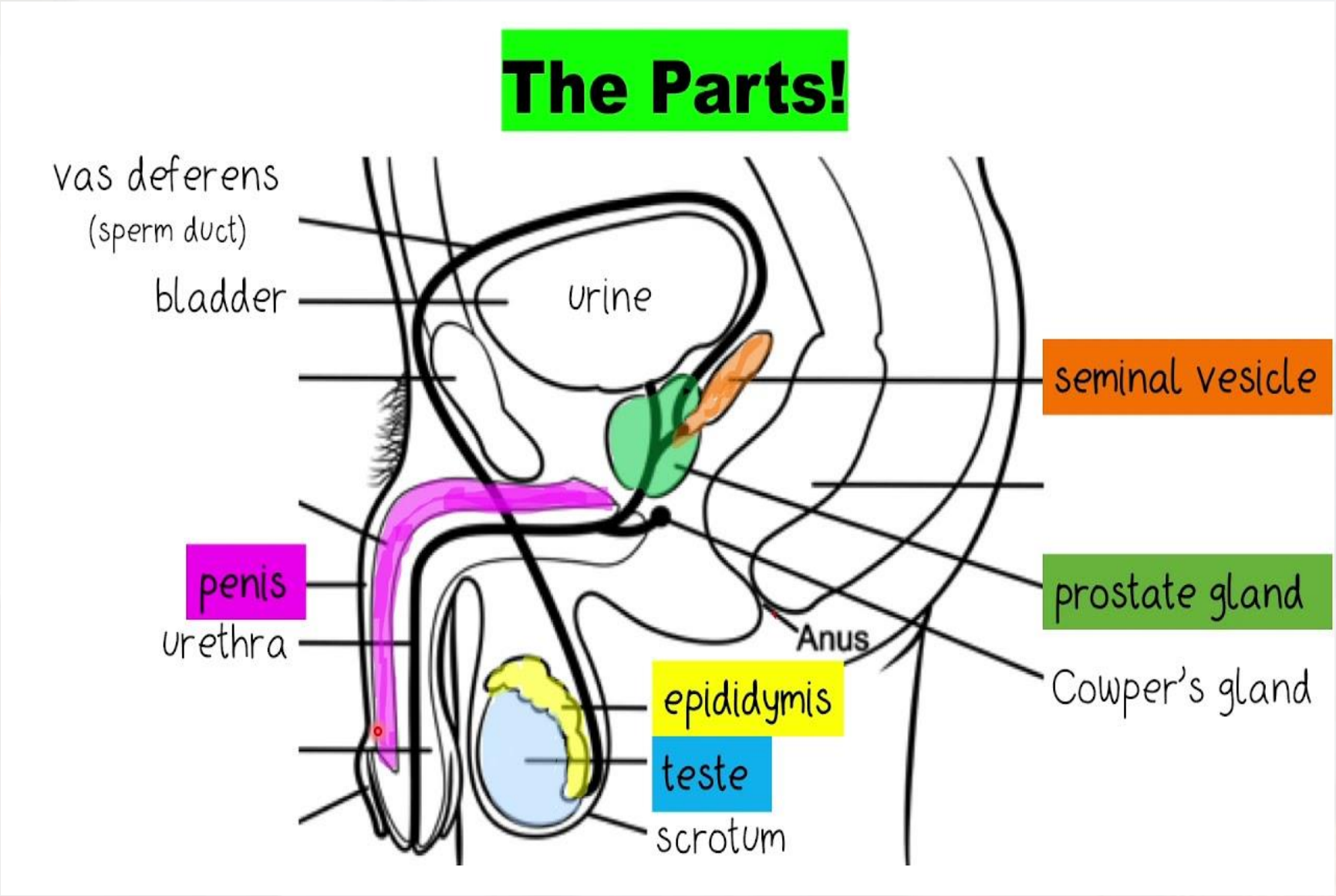


image from Ruth Hamilton, youtube video; the male reproductive system

Testicular Torsion – an Emergency

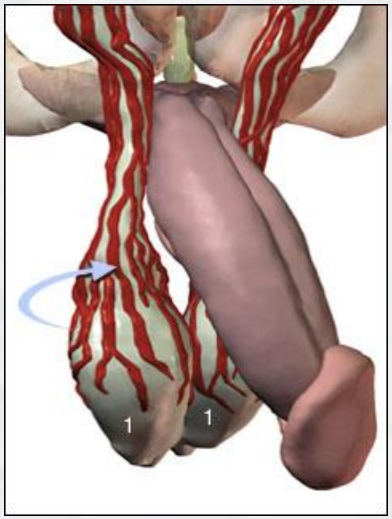
- Rare but urgent medical emergency
- spermatic cord that provides the blood supply to a testicle becomes twisted
- delay in diagnose and management can result in loss of a testicle
- Causes:
  - Congenital anomaly; bell clapper deformity; undescended testicle; sexual arousal and/or activity; trauma; exercise; cold weather
  - Important to note that it can happen during activity but exercise or other activity does not cause it; it can happen while simply standing or even sleeping
- Usually in males < 25
  - Most common in ages 12-18

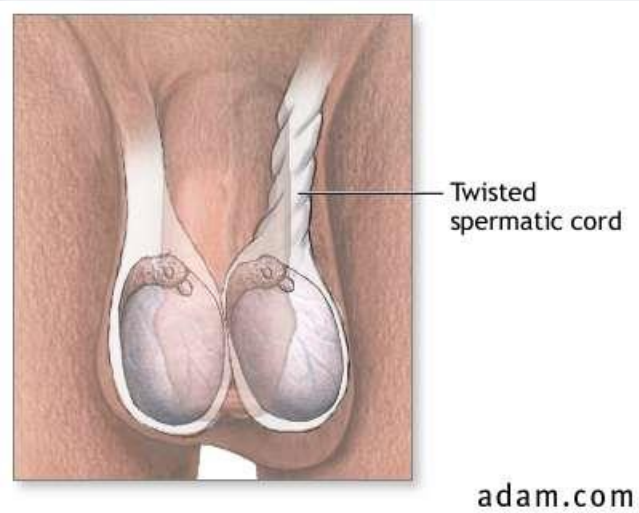
50% lose testcle after 6 hours of cut blood flow, 100% after 12 hours.

Testicular Torsion

- ***Usually sudden onset of severe unilateral scrotal pain***
- Note that the pain may also be in the lower abdominal quadrant or inguinal canal
- Pain may be constant or intermittent, but is not positional
- Hx: As many as 50% of patients have a history of prior episodes
- **REQUIRES SURGERY**

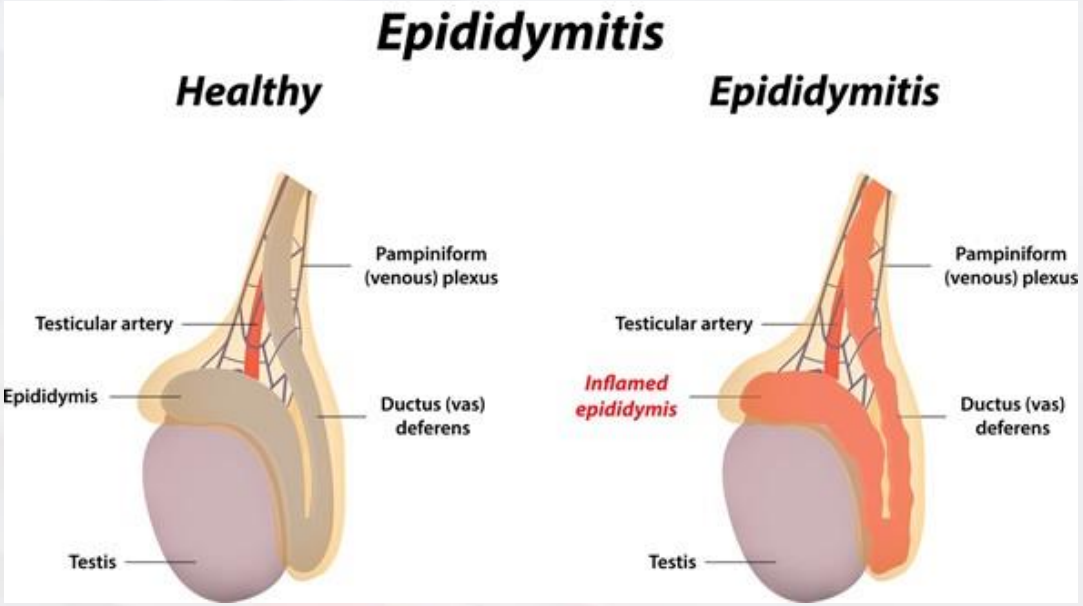
Testicular Torsion – an Emergency





Epididymitis and Orchitis

- inflammation or infection of the epididymis, a convoluted duct that lies on the posterior surface of the testicle.
- If the inflammation or infection extends to both the epididymis and the testicle, epididymo-orchitis is present.

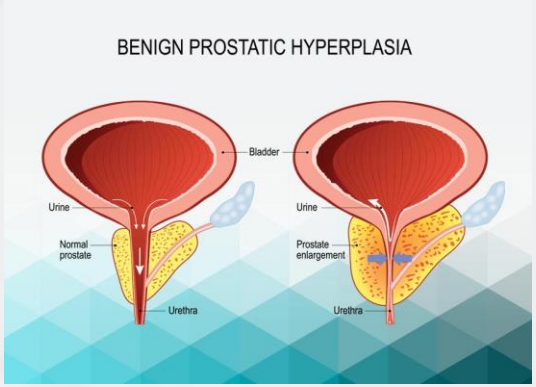


Epididymitis

- Hx:
  - Scrotal pain and edema
    - Usually gradual onset; Pain can be constant and severe
  - Recent surgery
  - STI
  - Urinary frequency, urgency, or dysuria (+/- Urethral discharge)
  - Urinary retention from bladder outlet obstruction in older patients
  - Nausea/ vomiting
  - Fever and chills
  - Abdominal or flank pain
  - Bilateral epididymal involvement (10%)

Prostate Enlargement

- Benign prostatic hyperplasia (BPH) is a noncancerous enlargement of the prostate gland that may restrict the flow of urine from the bladder
- prostate is a walnut-sized gland that forms part of the male reproductive system.
  - It increases in size as men age
- S&S
  - Urinary frequency, urgency, hesitancy
  - Incomplete bladder emptying
  - Straining, decrease force of stream, dribbling
- Rx: Phenoxybenzamine (Dibenzyline), Prazosin (Minipress), Alfuzosin (UroXatral), Indoramin, Terazosin (Hytrin), Doxazosin (Cardura), Tamsulosin (Flomax), ***Cialis*** (***Tadalafil***)***etc



Urinary Retention- an Emergency

- Common causes
  - Prostatic hypertrophy: benign or malignant
  - Anticholinergic drugs
  - bladder stabilizers and tricyclic antidepressants
  - Urethral calculus (stones)
  - Bladder neck hypertrophy
  - Prostatitis and prostatic abscess
  - UTI

Urinary Incontinence

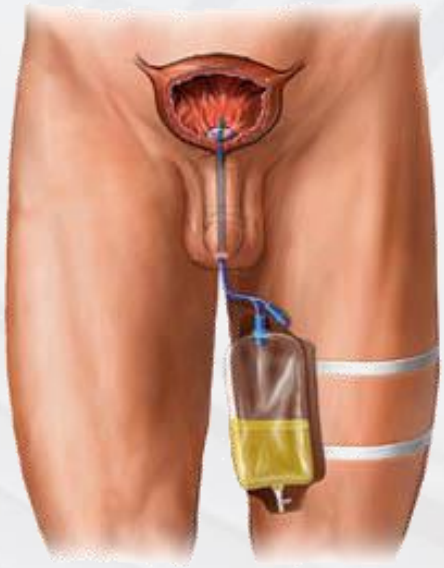
- involuntary release of urine
  - may be caused by physiologic, pharmacologic, pathologic, or psychological factors.
- Affects 15-30% of the population[1]
- Hx
  - Onset (pregnancy, postpartum, surgery, trauma)
  - Duration of complaint
  - Patterns (nocturnal vs diurnal)
  - Precipitants (cough, sneeze, position change, sound of running water)
  - Frequency/severity/quantity
    - Number of pads
    - Voiding diary?

Patient Urge to Urinate En Route

- Have them use the emesis bag

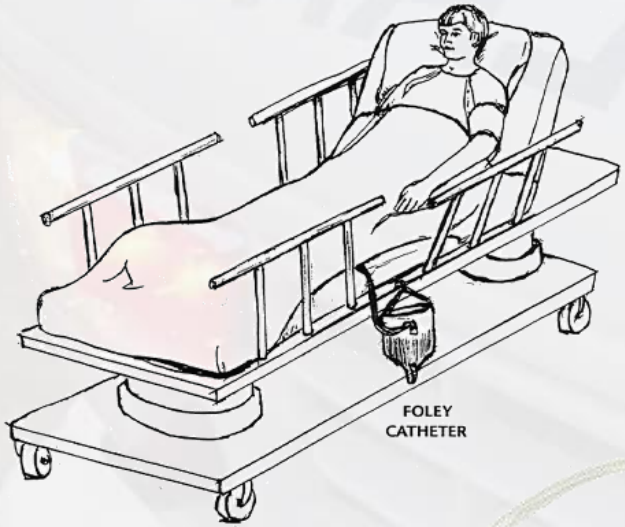
Foley Catheter Care

Foley catheter



- You may encounter some patients who are at home with a foley catheter – e.g. spinal injured patient
- You may be required to transfer a patient between facilities who has a foley
- If the foley tubing is long, secure it to the side of the stretcher below the bladder

Foley Catheter Care



- catheter is secured to the side of the stretcher - be very careful not to get the urinary bag or catheter caught on anything
  - As you move the stretcher or load it into the ambulance

Foley Catheter Care

- Wheelchair

Condom Catheter

- secured to the leg

Assessment

- OPQRST, AS-PN
- Fever? Chills? back/flank pain? suprapubic pain? Nausea? Vomiting?
- Sensation of bladder fullness
- Suprapubic pain
- Flank pain

Assessment

- Urgency?
- Frequency?
- Nocturia (getting up in the night to urinate)?
- Dysurea (difficulty urinating)?
- Gross hematurea? Seen in approx. 10%
- Oliguria, anurea? reduced vs none
- any changes in the color and/or consistency of the urine (e.g. purulent discharge)?

Assessment

- Ask elderly men about a history of prior UTI, prostate enlargement, urinary dribbling or hesitancy, or difficulty initiating the urinary stream
- history of indwelling catheters? other chronic urinary tract problems?
  - these patients are at much higher risk of UTI
- Peripheral edema (seen in AKI)

Assessment

- Be aware of risk of sepsis and septic shock in elderly males
  - Tachycardia
  - Hypotension
  - Tachypnea
  - fever

Assessment: Medication

Inquire about any antibiotic use

- Sulfamethoxazole-trimethoprim (Bactrim, Septra)
- Ciprofloxacin (Cipro)
- Ceftriaxone (Rocephin)
- Gentamicin (Garamycin, Gentacidin)
- Ertapenem (Invanz)
- Phenazopyridine (Pyridium, Urogesic)
- Cialis (Tadalafil)

Management

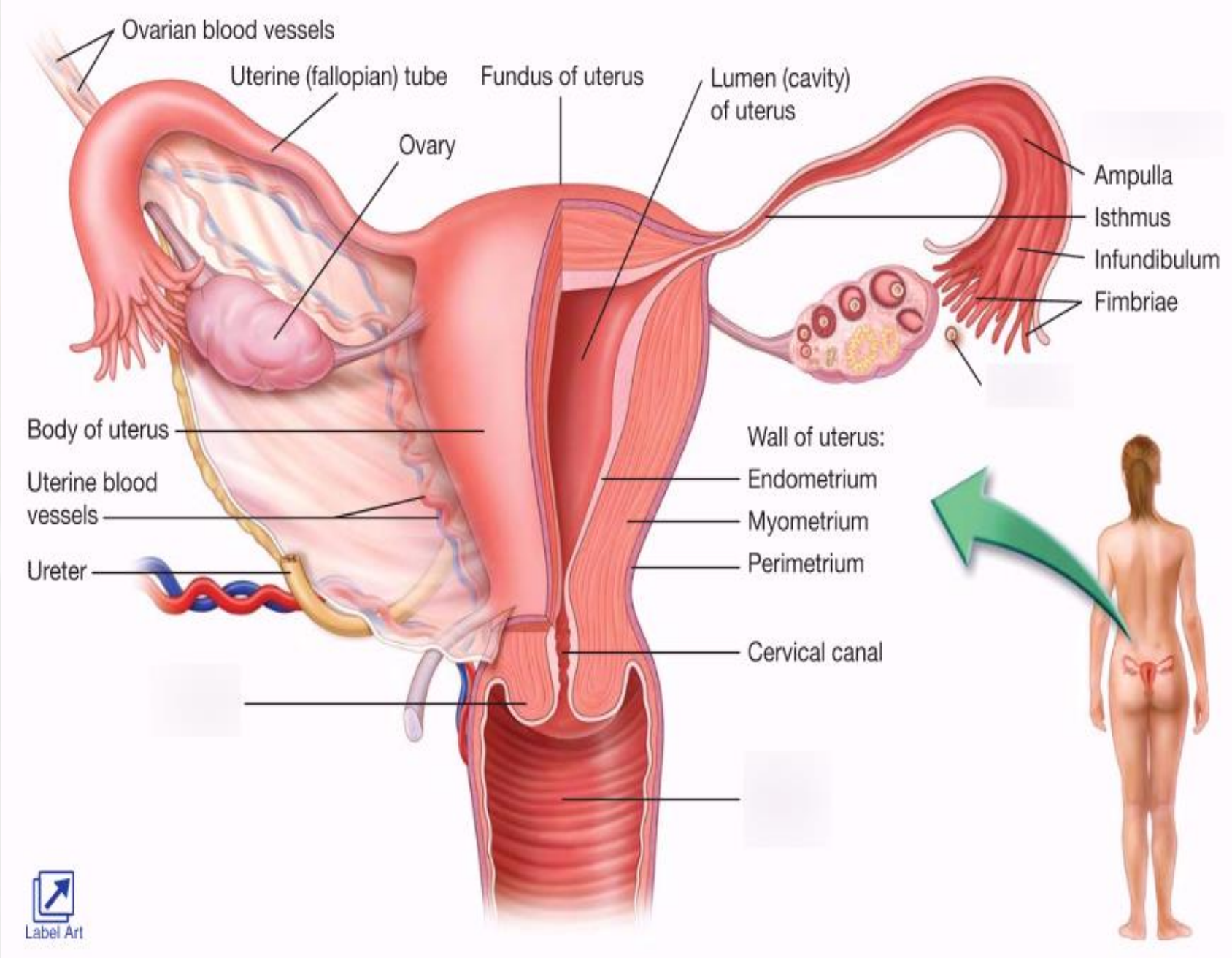
- The vast majority of patients with GU complaints make their own way to the E.R.
- A 9-1-1 call should raise suspicions (based on circumstance)

Management

Supportive in nature

- ECG, SpO_2
- O_2 prn
- IV TKO prn
- Findings of hypotension, tachycardia, and delayed capillary refill may signify volume depletion/shock
  - This requires intravenous (IV) fluid resuscitation in the field
- Consider ketorolac for analgesia (see ALS PCS)
- Temperature
- Prepare the stretcher for incontinence
- Take care of Foley catheter
- Transport in position of comfort

Female Reproductive Disorders


Ovarian blood vessels
Uterine (fallopian) tube
Fundus of uterus
Lumen (cavity) of uterus
Ovary
Ampulla
Isthmus
Infundibulum
Fimbriae
Body of uterus
Uterine blood vessels
Ureter
Wall of uterus:
Endometrium
Myometrium
Perimetrium
Cervical canal
Label Art

Dysmenorrhea/amenorrhea

- Dysmenorrhea – painful uterine cramping
- Amenorrhea – loss of normal menstruation (3 missed periods in a row OR girls who have not begun menstruating by age 15
- Menorrhagia – excessive menstrual bleeding give TXA
  - Can happen at any age – more common in teenagers or perimenopause
- Most common cause of non-traumatic vaginal bleeding is a spontaneous abortion (miscarriage)
- Management:
  - Do not assume vaginal bleeding is from normal menstruation
  - Supportive Care – Fluid resuscitation as needed

Gynecological

- PID – Pelvic Inflammatory Disease
  - An infection of the female reproductive tract (most commonly uterus, fallopian tubes, ovaries but can include peritoneum and intestines)
  - Disease causing organisms enter via the vagina
    - Most common causes – gonorrhea, chlamydia
    - Less common – streptococcal, staphylococcal
  - Most commonly in sexually active individuals

Gynecological

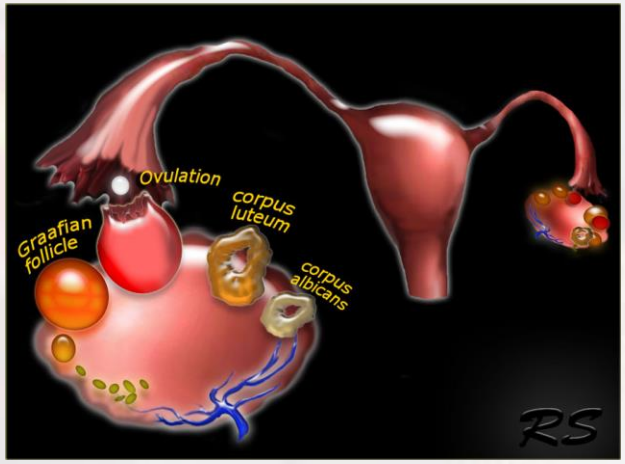
- PID – Pelvic Inflammatory Disease
  - Presentation;
    - Diffuse pain to lower abdomen
    - Moderate to severe (may be difficult to distinguish from appendicitis)
    - Increased pain with palpation, sexual intercourse, movement (walk with a shuffling gait)
  - Management
    - Supportive care
    - Pain management

Endometriosis/Endometritis

- Endometritis – infection of the uterine lining
  - Causes –complication of miscarriage, childbirth or gynecological procedures
- Endometriosis –
  - Condition where endometrial tissue is found outside the uterus – this tissue responds to the hormonal changes of the menstrual cycle (bleeding)
  - Common with infertility

Ovarian Cysts

- Fluid filled pockets that develop in the ovaries. When thy rupture they can cause severe abdominal pain
- Often created when the egg leaves the ovary*
- When ruptured, a small amount of blood leaks into the abdomen (irritating to the peritoneal lining)
  - What type of pain will the patient present with? (assessment finding)



Interstitial Cystitis

- Most commonly seen in females (~90%)
- Characterized as a chronic bladder condition
  - Unknown etiology
  - Chronically inflamed or irritated bladder wall
  - Range in pain from mild to severe
    - Often increases during sexual activity or menstruation
  - Symptoms mimic that of UTI or STI's
- Management is supportive and symptom relief
  - No cure
  - Antibiotics are ineffective/ not required

QUESTIONS

References

1. Ogundele O, Silverberg M (2006: Urinary Incontenence. eMedicine, http://www.emedicine.com/emerg/topic791.htm retrieved Feb 4, 2007
2. Caroline, N. (2021). Emergency Care in the Streets (Russel D. MacDonald, ED.). Jones and Bartlett